

Study Questions: Cognition

1. When you read a crochet pattern before beginning a project, what cognitive process is occurring in Active Inference terms? How does this differ from actually crocheting the pattern?
2. Describe the experience of mental stitch counting. How does holding a number in working memory while stitching illustrate the concept of internal simulation?
3. Many experienced crocheters can crochet while watching television or carrying on a conversation. Explain this ability using the concept of distributed generative models and embodied cognition.
4. A beginner reads "ch 3, skip 2, dc in next st" and must perform each step deliberately. An expert reads the same instruction and immediately visualizes a filet mesh. What has changed in the expert's generative model to enable this?
5. When a crocheter wonders "Will this be long enough if I add one more repeat?" they are engaging in counterfactual reasoning. Describe this process in Active Inference terms.
6. You are working a complex cable stitch pattern and suddenly the television shows something surprising. You lose your place in the pattern. Explain this in terms of cognitive resource allocation and prediction error propagation.
7. How does the depth of a generative model relate to the crocheter's ability to predict how a finished garment will drape? What knowledge is encoded in a deep model versus a shallow one?
8. When a crocheter modifies a pattern in real time — adding extra stitches, changing a repeat count — what kind of cognitive simulation are they performing?
9. Describe the shift from automatic to controlled processing that occurs when a crocheter encounters an unexpected instruction or an unfamiliar stitch. What triggers this shift in the Active Inference framework?

10. A crocheter holds a conversation while working a simple row of single crochet, but must stop talking when they reach a tricky increase section. What does this reveal about the cognitive demands of different stitch operations?
11. How does pattern reading in crochet compare to reading sheet music? What role does the generative model play in both cases?
12. When you chain a specific number at the start of a project, you count each chain. Is this counting an example of perception, cognition, or both? Explain using Active Inference concepts.
13. Some crocheters describe "getting lost in thought" while crocheting repetitive rows, only to realize they have completed several inches without conscious awareness. What level of the generative model hierarchy was active during this period?
14. A crocheter decides to substitute a double crochet for a half double crochet throughout a pattern. What cognitive simulation must they perform to predict how this substitution will affect the final dimensions?
15. How does the concept of working memory capacity relate to the complexity of crochet patterns a person can manage? Why might a very complex pattern require external aids (charts, row counters, stitch markers)?
16. When you look at a ball of yarn and imagine it as a finished scarf, you are running a forward simulation. What parameters of the generative model contribute to this visualization?
17. Describe how the hierarchical structure of a crochet pattern (project level, section level, row level, stitch level) maps onto the hierarchical structure of the generative model in Active Inference.
18. A crocheter who has never made a three-dimensional amigurumi toy attempts one for the first time. How does their generative model handle the transition from flat 2D fabric cognition to 3D construction?

19. When you make a mistake and realize it three rows later, you must mentally reconstruct what happened. Describe this retrospective cognitive process in terms of the generative model inferring hidden past states from current observations.
20. Reflect on your own experience of "thinking with your hands" in crochet. At what point in your learning journey did your hands start to "know" the stitches? What changed cognitively when this transition occurred?